

✓ CS568 Deep Learning

Department of Computer Science
University of the Punjab

Recitation 1: Basics of Python

✓ Python

Python is a high-level, open source and interpreted programming language. It is almost like pseudo-code, allows you to express very powerful ideas in few lines of code. Why Python?

- Simple and easy to learn
- Used to build web applications
- Used with big data frameworks
- Extensive support libraries
- Must-know language for deep learning

In this recitation, we will discuss

1. Data types, operators and built-in functions
2. Data containers (lists, tuple, dictionary and sets)
3. Statements and loops
4. Functions and classes

i) Take input from user

Start coding or [generate](#) with AI.

```
name = input("What's your name? ")  
print('Hi, ', name)
```

```
What's your name? ayesha  
Hi, ayesha
```

✓ ii) Data types

```
a = 1 # integer  
print(type(a))
```

```
<class 'int'>
```

```
a = 1.1 # float  
print(type(a))
```

```
<class 'float'>
```

```
a = "hello" # string using double quotes  
print(type(a))  
b = 'hello' # string using single quote  
print(type(b))
```

```
<class 'str'>  
<class 'str'>
```

Python is a strongly typed language, variables are bound to specific data types. For example

```
value = 10 + int("10")  
print(value)
```

```
20
```

```
number = input("Please enter any number between 1 and 10: ")
value = 10 + int(number)
print(value)
```

```
Please enter any number between 1 and 10: 5
15
```

```
number = input("Please enter any number between 1 and 10: ")
value = 10 + int(number)
print(value)
```

```
Please enter any number between 1 and 10: 15
25
```

iii) Type conversion

- `int()` to convert a string into an integer.
- `float()` to convert a string into a float.
- `bool()` to convert a string into a boolean value.
- To check the type of variable use `print(type(number))`

iv) Strings

```
course = 'Deep Learning'
print(len(course)) # to get the length of string
```

```
13
```

```
combine_s = 'pak'+ 'istan'
print(combine_s) # combine two strings using + operator
```

```
pakistan
```

```
repeat_d = '#'*15 # repeat string multiple times using * operator
print(repeat_d)
```

```
#####
```

```
course = 'Deep Learning' # square brackets used to access the elements of string
print(course[0:4])      #JUST DISPLAY Deep # Fill in the blank
```

```
Deep
```

```
course = 'DEEP LEARNING'
print(course.lower()) # lower() method returns the string in lower case
```

```
deep learning
```

```
course = 'deep learning'
print(course.upper()) # upper() method returns the string in upper case
```

```
DEEP LEARNING
```

```
height = 'tall'
print(height.replace('t','m')) # replace() method used to swap elements of string
```

```
mall
```

```
splitted = course.split(' ') # split the string into two sub-strings
print(splitted)
```

```
['deep', 'learning']
```

```
check_var = 'deep' in course # check if the phrase is in the string or not
print(check_var)
```

```
True
```

```
acc = 90
txt = "testing {}% accuracy"
print(txt.format(acc)) # combine text with numbers
```

```
testing 90% accuracy
```

```
acc = 90.845689
txt = "testing accuracy: {:.2f}%"
print(txt.format(acc)) # combine text with numbers with two decimal format
```

```
testing accuracy: 90.85%
```

v) Common operators in python

```
x = 5
y = 2
print("x+y = ",x + y) # addition
print("x-y = ",x-y) # subtraction
print("x*y = ",x * y) # multiplication
print("x/y = ",x / y) # division
print("x%y = ",x % y) # modulus #FILL IN THE BLANK
print("x**y = ",x**y) # power
```

```
x+y = 7
x-y = 3
x*y = 10
x/y = 2.5
x%y = 1
x**y = 25
```

✓ vi) Logical operators to combine conditional statements

```
x = 5
y = 5
print(x > 3 and x < 10)
print(x < 5 or x < 4) ##FILL IN THE BLANK
print(not(x > 3 and x < 10))
```

```
True
False
False
```